Elementary Problems
Inequality

Version 1.0.1

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Basic Work

1. Let x be a real number. Find the minimum value of $4x^2 - 12x + 27$.
2. Let x and y be real numbers. Find the minimum value of $x^2 - 4xy + 5y^2 - 2y$.
3. Let x and y be positive real numbers such that $x \geqslant y$. Prove that
$$x - y \geqslant \frac{1}{x} - \frac{1}{y}.$$
4. Let x and y be real numbers such that $x \geqslant y$. Prove that
$$x^3 + x^2y \geqslant y^3 + xy^2.$$
5. Let x and y be positive real numbers. Prove that
$$\frac{x^2}{y} + \frac{y^2}{x} \geqslant x + y.$$
6. Let x be a real number satisfying $0 < x < 1$. Prove that
$$\sqrt{x} + \sqrt{1-x} \leqslant \sqrt{\frac{5}{6}(3-x)}.$$
7. Let x and y be positive real numbers. Prove that
$$\frac{x}{x^2+4} + \frac{y}{y^2+4} \leqslant \frac{1}{2}.$$
8. (IMO 1965 Problem 2) Consider the system of equations
$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= 0, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= 0, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= 0 \end{aligned}$$
with unknowns x_1, x_2, x_3. The coefficients satisfy the conditions:
(a) a_{11}, a_{22}, a_{33} are positive numbers;
(b) the remaining coefficients are negative numbers;
(c) in each equation, the sum of the coefficients is positive.
Prove that the given system has only the solution $x_1 = x_2 = x_3 = 0$.

9. (IMO shortlist 1981) Find the minimum value of

$$\max\{a+b+c,\ b+c+d,\ c+d+e,\ d+e+f,\ e+f+g\}$$

subject to the constraints

(i) $a,b,c,d,e,f,g \geqslant 0$,

(ii) $a+b+c+d+e+f+g=1$.

10. (IMO shortlist 1971) Prove the inequality

$$\frac{a_1+a_3}{a_1+a_2}+\frac{a_2+a_4}{a_2+a_3}+\frac{a_3+a_1}{a_3+a_4}+\frac{a_4+a_2}{a_4+a_1} \geqslant 4,$$

where $a_i > 0$, $i=1,2,3,4$.

11. (IMO 1974 Problem 5) Determine all possible values of

$$S=\frac{a}{a+b+d}+\frac{b}{a+b+c}+\frac{c}{b+c+d}+\frac{d}{a+c+d}$$

where a,b,c,d are arbitrary positive numbers.

12. (IMO 1971 Problem 1) Prove that the following assertion is true for $n=3$ and $n=5$, and that it is false for every other natural number $n>2$:

If $a_1,a_2,\ldots,a_n$ are arbitrary real numbers, then

$$(a_1-a_2)(a_1-a_3)\cdots(a_1-a_n)+(a_2-a_1)(a_2-a_3)\cdots(a_2-a_n)$$
$$+\cdots+(a_n-a_1)(a_n-a_2)\cdots(a_n-a_{n-1}) \geqslant 0.$$

13. Let x and y be positive real numbers such that $x+y=2$. Prove that

$$\sqrt{2x^2+2y^2}+4\sqrt{x}+4\sqrt{y} \leqslant 10.$$

14. (IMO 1967 Problem 5) Consider the sequence $\{c_n\}$, where

$$\begin{aligned} c_1 &= a_1+a_2+\cdots+a_8,\\ c_2 &= a_1^2+a_2^2+\cdots+a_8^2,\\ &\ldots,\\ c_n &= a_1^n+a_2^n+\cdots+a_8^n,\\ &\cdots \end{aligned}$$

in which $a_1,a_2,\ldots,a_8$ are real numbers not all equal to zero. Suppose that an infinite number of terms of the sequence $\{c_n\}$ are equal to zero. Find all natural numbers n for which $c_n=0$.

AM-GM Inequality

15. Let a, b, c be positive real numbers. Prove that

$$\frac{a}{b}+\frac{b}{c}+\frac{c}{a} \geqslant 3.$$

16. Let x be a positive real number. Prove that

$$x^3+x+\frac{4}{x} \geqslant 6.$$

17. Let a, b, c be positive real numbers. Prove that

$$a^2+b^2+c^2 \geqslant ab+bc+ca.$$

18. Let a, b, c be positive real numbers. Prove that

$$(a+b)(a+c) \geqslant 2\sqrt{abc(a+b+c)}.$$

19. (IMO shortlist 1966)

Given n positive real numbers $a_1, a_2, \ldots, a_n$ such that $a_1 a_2 \cdots a_n = 1$, prove that

$$(1+a_1)(1+a_2)\cdots(1+a_n) \geqslant 2^n.$$

20. Let x, y, z be positive real numbers such that $xyz = 8$. Prove that

$$(x^2+y)(y^2+z)(z^2+x) \geqslant 216.$$

21. Let n be a positive integer and let $a_1, a_2, \ldots, a_n$ be positive real numbers. Suppose $b_1, b_2, \ldots, b_n$ is a permutation of $a_1, a_2, \ldots, a_n$. Find the minimum value of

$$\left(a_1^2+\frac{1}{b_1^2}\right)\left(a_2^2+\frac{1}{b_2^2}\right)\cdots\left(a_n^2+\frac{1}{b_n^2}\right)$$

by considering all possible a_j's.

22. (IMO shortlist 1968) If a and b are arbitrary positive real numbers and m is an integer, prove that

$$\left(1+\frac{a}{b}\right)^m+\left(1+\frac{b}{a}\right)^m \geqslant 2^{m+1}.$$

23. Let a, b, c be positive real numbers satisfying $a+b+c=1$. Find the maximum value of a^3b^2c.

24. Let a, b, c be positive real numbers satisfying $a+b+c=3$. Prove that

$$a^2b+b^2c+c^2a \leqslant a^2+b^2+c^2.$$

25. (IMO shortlist 1967) Solve the system of equations:

$$\begin{aligned} x^2+x-1&=y,\\ y^2+y-1&=z,\\ z^2+z-1&=x. \end{aligned}$$

Cauchy-Schwarz Inequality

26. Find the optimum values of $\sin\theta+\cos\theta$ where θ is a real number.

27. Let a, b, c, x, y, z be positive real numbers. Prove that

$$\frac{a^2}{x}+\frac{b^2}{y}+\frac{c^2}{z} \geqslant \frac{(a+b+c)^2}{x+y+z}.$$

28. Let a, b, c, d be positive real numbers. Prove that

$$\frac{1}{a}+\frac{1}{b}+\frac{4}{c}+\frac{16}{d} \geqslant \frac{64}{a+b+c+d}.$$

29. Let a, b, c be positive real numbers. Prove that

$$2\sqrt{a+2b+4c} \geqslant \sqrt{a}+2\sqrt{b}+2\sqrt{c}.$$

30. Let a, b, c be positive real numbers such that $a>c$ and $b>c$. Prove that

$$\sqrt{c(a-c)}+\sqrt{c(b-c)} \leqslant \sqrt{ab}.$$

31. (USAMO 1978 Problem 1) Given that a, b, c, d, e are real numbers such that

$$\begin{aligned} a+b+c+d+e&=8,\\ a^2+b^2+c^2+d^2+e^2&=16, \end{aligned}$$

determine the maximum value of e.

32. (IMO shortlist 1993 A3) Prove that

$$\frac{a}{b+2c+3d}+\frac{b}{c+2d+3a}+\frac{c}{d+2a+3b}+\frac{d}{a+2b+3c} \geqslant \frac{2}{3}$$

for all positive real numbers a, b, c, d.

33. (IMO shortlist 1990) Let w, x, y, z be nonnegative reals such that $wx+xy+yz+zw=1$. Show that
$$\frac{w^3}{x+y+z}+\frac{x^3}{w+y+z}+\frac{y^3}{w+x+z}+\frac{z^3}{w+x+y} \geqslant \frac{1}{3}.$$

34. (IMO 1995 Problem 2) Let a, b, c be positive real numbers such that $abc = 1$. Prove that
$$\frac{1}{a^3(b+c)}+\frac{1}{b^3(c+a)}+\frac{1}{c^3(a+b)} \geqslant \frac{3}{2}.$$

35. (IMO 1982 Problem 3) Consider the infinite sequences $\{x_n\}$ of positive real numbers with the following properties:
$$x_0 = 1, \text{and for all } i \geqslant 0,\ x_{i+1} \leqslant x_i.$$
   (a) Prove that for every such sequence, there is an $n \geqslant 1$ such that
$$\frac{x_0^2}{x_1}+\frac{x_1^2}{x_2}+\cdots+\frac{x_{n-1}^2}{x_n} \geqslant 3.999.$$
   (b) Find such a sequence such that for all n:
$$\frac{x_0^2}{x_1}+\frac{x_1^2}{x_2}+\cdots+\frac{x_{n-1}^2}{x_n} < 4.$$

36. (IMO shortlist 1987) Show that if a, b, c are the lengths of the sides of a triangle and if $2S = a+b+c$, then
$$\frac{a^n}{b+c}+\frac{b^n}{c+a}+\frac{c^n}{a+b} \geqslant \left(\frac{2}{3}\right)^{n-2} S^{n-1} \quad \forall n \in \mathbb{N}.$$

Rearrangement Inequality, Chebyshev's Inequality

37. Let a, b, c be positive real numbers. Prove that
$$a^3+b^3+c^3 \geqslant a^2b+b^2c+c^2a.$$

38. (IMO 1975 Problem 1) Let x_i, y_i $(i = 1, 2, \ldots, n)$ be real numbers such that
$$x_1 \geqslant x_2 \geqslant \cdots \geqslant x_n \quad \text{and} \quad y_1 \geqslant y_2 \geqslant \cdots \geqslant y_n.$$
Prove that, if $z_1, z_2, \ldots, z_n$ is any permutation of $y_1, y_2, \ldots, y_n$, then
$$\sum_{i=1}^{n}(x_i-y_i)^2 \leqslant \sum_{i=1}^{n}(x_i-z_i)^2.$$

39. (IMO 1978 Problem 5) Let $\{a_k\}$ $(k = 1, 2, 3, \ldots, n, \ldots)$ be a sequence of distinct positive integers. Prove that for all natural numbers n,

$$\sum_{k=1}^{n} \frac{a_k}{k^2} \geqslant \sum_{k=1}^{n} \frac{1}{k}.$$

40. Let n be a positive integer and let $a_1, a_2, \ldots, a_n$ be pairwise distinct positive integers. Prove that

$$\frac{a_1}{1 \cdot 2^1} + \frac{a_2}{2 \cdot 2^2} + \frac{a_3}{3 \cdot 2^3} + \cdots + \frac{a_n}{n \cdot 2^n} \geqslant 1 - \frac{1}{2^n}.$$

41. Let a, b, c be positive real numbers. Prove that

$$\frac{a^2 + bc}{b + c} + \frac{b^2 + ca}{c + a} + \frac{c^2 + ab}{a + b} \geqslant a + b + c.$$

42. Let a, b, c be positive real numbers. Prove that

$$\frac{(b+c)(b^2+c^2)}{a} + \frac{(c+a)(c^2+a^2)}{b} + \frac{(a+b)(a^2+b^2)}{c} \geqslant 4(a^2+b^2+c^2).$$

43. (USAMO 1974 Problem 2) Prove that if a, b, and c are positive real numbers, then

$$a^a b^b c^c \geqslant (abc)^{\frac{a+b+c}{3}}.$$

Muirhead's Theorem, Schur's Inequality

44. Let a, b, c be positive real numbers. Prove that

$$2abc(a^3 + b^3 + c^3) \geqslant a^3b^2c + a^3bc^2 + a^2b^3c + a^2bc^3 + ab^3c^2 + ab^2c^3.$$

45. Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$a^2 + b^2 + c^2 \geqslant a + b + c.$$

46. (India Regional MO 1994 Problem 8)

If a, b, c are positive real numbers such that $a + b + c = 1$, prove that

$$(1+a)(1+b)(1+c) \geqslant 8(1-a)(1-b)(1-c).$$

47. Let a, b, c be nonnegative real numbers satisfying $a + b + c = 1$. Prove that

$$a^3 + b^3 + c^3 + 6abc \geqslant \frac{1}{4}.$$

48. Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that
$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} + 3 \geqslant a + b + c + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

49. Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that
$$a^2 + b^2 + c^2 + 3 \geqslant 2(ab + bc + ca).$$

50. (IMO 1984 Problem 1) Prove that $0 \leqslant yz + zx + xy - 2xyz \leqslant \frac{7}{27}$, where x, y and z are nonnegative real numbers for which $x + y + z = 1$.

Jensen's Inequality, Majorization Inequality

51. Let $x_1, x_2, \ldots, x_n$ be positive real numbers satisfying $x_1 + x_2 + \cdots + x_n = 1$. Prove that
$$\frac{x_1}{2 - x_1} + \frac{x_2}{2 - x_2} + \cdots + \frac{x_n}{2 - x_n} \geqslant \frac{n}{2n - 1}.$$

52. Let a, b, c, d be positive real numbers satisfying $a + b + c + d = 2$. Prove that
$$\left(\frac{a}{a+2}\right)^3 + \left(\frac{b}{b+2}\right)^3 + \left(\frac{c}{c+2}\right)^3 + \left(\frac{d}{d+2}\right)^3 \geqslant \frac{4}{125}.$$

53. Let a, b, c, d be nonnegative real numbers satisfying $a+b+c+d = \frac{1}{2}$. Find the optimum values of
$$\frac{4}{a+1} + \frac{4}{b+1} + \frac{4}{c+1} + \frac{4}{d+1} - (a^2 + b^2 + c^2 + d^2).$$

54. (IMO 1961 Problem 3) Solve the equation $\cos^n x - \sin^n x = 1$, where n is a natural number.

Local Refinement Method

55. Find the three-digit number n for which the quotient of n and the sum of digits of n is the smallest.

56. Let a and b be nonnegative real numbers satisfying $a + b \leqslant 1$. Find the maximum value of $3a^2 + 2ab + b$.

57. (IMO 1976 Problem 4) Determine, with proof, the largest number which is the product of positive integers whose sum is 1976.

58. (CMO 1989 Problem 5) There are 1989 points in the space such that any three points are not collinear. We divide the points into 30 groups of pairwise distinct sizes. We choose a point each from three different groups to form a triangle. What should the numbers of points of the groups be in order to maximize the number of such triangles?

59. (IMO shortlist 1974) The sum of the squares of five real numbers a_1, a_2, a_3, a_4, a_5 equals 1. Prove that the least of the numbers $(a_i - a_j)^2$, where $i, j = 1, 2, 3, 4, 5$ and $i \neq j$, does not exceed $\frac{1}{10}$.

Geometric Inequality (Algebraic)

60. Let a, b, c be the lengths of a triangle. Prove that

$$2(a^3 + b^3 + c^3) < (a + b + c)(a^2 + b^2 + c^2).$$

61. Let a, b, c be the lengths of a triangle. Prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < 2.$$

62. (IMO 1983 Problem 6) Let a, b and c be the lengths of the sides of a triangle. Prove that

$$a^2b(a-b) + b^2c(b-c) + c^2a(c-a) \geqslant 0.$$

Determine when equality occurs.

63. (IMO 1961 Problem 2) Let a, b, c be the sides of a triangle, and T its area. Prove:

$$a^2 + b^2 + c^2 \geqslant 4\sqrt{3}T.$$

In what case does equality hold?

64. (IMO 1981 Problem 1) P is a point inside a given triangle ABC. D, E, F are the feet of the perpendiculars from P to the lines BC, CA, AB respectively. Find all P for which

$$\frac{BC}{PD} + \frac{CA}{PE} + \frac{AB}{PF}$$

is least.

Induction

65. Prove that $3^{n-1} \geqslant 2n-1$ for any positive integer n.

66. Prove that $n^n \geqslant (n+1)^{n-1}$ for any positive integer n.

67. (IMO shortlist 1967) Prove that

$$\frac{1}{3}n^2 + \frac{1}{2}n + \frac{1}{6} \geqslant (n!)^{\frac{2}{n}},$$

and let $n \geqslant 1$ be an integer. Prove that this inequality is only possible in the case $n = 1$.

68. (IMO shortlist 1966) Let n be a positive integer. Prove that:

(a) $\log_{10}(n+1) > \dfrac{3}{10n} + \log_{10} n$;

(b) $\log_{10} n! > \dfrac{3n}{10}\Big(\dfrac{1}{2} + \dfrac{1}{3} + \cdots + \dfrac{1}{n} - 1\Big)$.

69. (Poland MO Finals 2001 Problem 1) Prove the following inequality:

$$x_1 + 2x_2 + 3x_3 + \cdots + nx_n \leqslant \frac{n(n-1)}{2} + x_1 + x_2^2 + x_3^3 + \cdots + x_n^n$$

where $x_i > 0$ for all i.

70. Let $a_1, a_2, \ldots, a_n$ be positive real numbers. Show that

$$a_1^2 + a_1a_2^2 + a_1a_2a_3^2 + \cdots + a_1a_2\cdots a_{n-1}a_n^2 \geqslant 4a_1a_2\cdots a_n - 4.$$

71. (IMO shortlist 1979) Consider the sequences $(a_n), (b_n)$ defined by

$$a_1 = 3, \quad b_1 = 100, \quad a_{n+1} = 3^{a_n}, \quad b_{n+1} = 100^{b_n}.$$

Find the smallest integer m for which $b_m > a_{100}$.

72. (IMO shortlist 1966) Let $a_1, a_2, \ldots, a_n$ be positive real numbers. Prove the inequality

$$\binom{n}{2}\sum_{i<j}\frac{1}{a_ia_j} \geqslant 4\Big(\sum_{i<j}\frac{1}{a_i+a_j}\Big)^2.$$

Record

v1.0: problems 1–72